Translationese All the Way Down

Auditing what "supports Twi" means in social media NLP resources

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DCIT 316: Computational Models for Social Media Mining · University of Ghana

https://github.com/osamabinlaggin/dcit316-final-project

The claims

Twi is supported, according to the people who built the tools

Google Translate added Twi and Ewe in 2022.

Meta's NLLB-200 covers both. Its benchmark, FLORES-200, lists twi_Latn, aka_Latn and ewe_Latn.

AfriSenti, AfriHate and MasakhaNER 2.0 all list Twi.

From Accra, it looks as if the tools exist. This project checks what they do with Twi as people write it online.

Three research questions

RQ1 (accountability). What do the resources that claim Twi disclose about which Twi, where the text came from, who labelled it and how much they agreed?

RQ2 (distribution). How far is the text those claims are measured on from Twi on social media?

RQ3 (capability). How do the tools a Ghanaian analyst would reach for perform on natively written Twi tweets, against a baseline trained on Twi?

Finding 1 · coverage

Ga, the language of Accra, is in none of it

Language	FLORES-200	MasakhaNER 2.0	AfriSenti	AfriHate	Google Tr.	NLLB-200
Twi / Akan	yes (2 files)	yes	yes	yes	yes	yes
Ewe	yes	yes	no	no	yes	yes
Ga	no	no	no	no	no	no

Ewe has benchmarks but no social media data at all. Ga has nothing to evaluate on and nothing to train on.

Finding 2 · provenance

The Twi benchmarks are translations into Twi

Corpus	Natively written?	Tokens using ɛ/ɔ	Texts writing 3 for ɛ	Mean tokens
FLORES-200 twi_Latn	No, translated English Wikipedia	28.5%	0.1%	25.3
MasakhaNER 2.0 twi	No, MAFAND-MT translated news	34.6%	0.0%	23.4
AfriSenti twi (tweets)	Yes	5.6%	42.5%	9.3

FLORES Twi vocabulary covers 15% of the word types in the tweets.

FLORES aka_Latn and twi_Latn are separate translations: 8 of 2009 lines identical. Neither says which dialect it is.

Finding 3 · accountability

What the documentation lets you check

Criterion	FLORES-200	AfriSenti	AfriHate	MasakhaNER 2.0
A1 Language referent precise	no dialect named	no dialect named	no dialect named	merges 'Akan/Twi'
A2 Natively written	translated (Wikipedia)	YES - tweets	YES - tweets	translated (MAFAND-MT)
A3 Native-speaker annotators stated	human-verified translation	yes	yes, explicit	yes
A4 Agreement reported	n/a	yes, k=0.51	yes, 0.46-0.81	yes, k=0.932 POST-ADJUDICATION
A5 Compensation disclosed	not found	not found	yes - 'more than the minimum wage'	yes - all annotators remunerated
A6 Datasheet / data statement	not found	not found	not found	not found

No resource ships a datasheet. MasakhaNER's kappa is measured after adjudication; AfriSenti's is not, so the two numbers cannot be compared.

RQ3 · design

Method: five systems, one baseline, 949 test tweets

Data: AfriSenti Twi. Train 3,481, dev 388, test 949. C tuned on dev; test used once.

Baseline: TF-IDF (character 2-5 grams + words) with logistic regression, trained on Twi.

English-only: Twitter-RoBERTa, the tool an analyst reaches for by default.

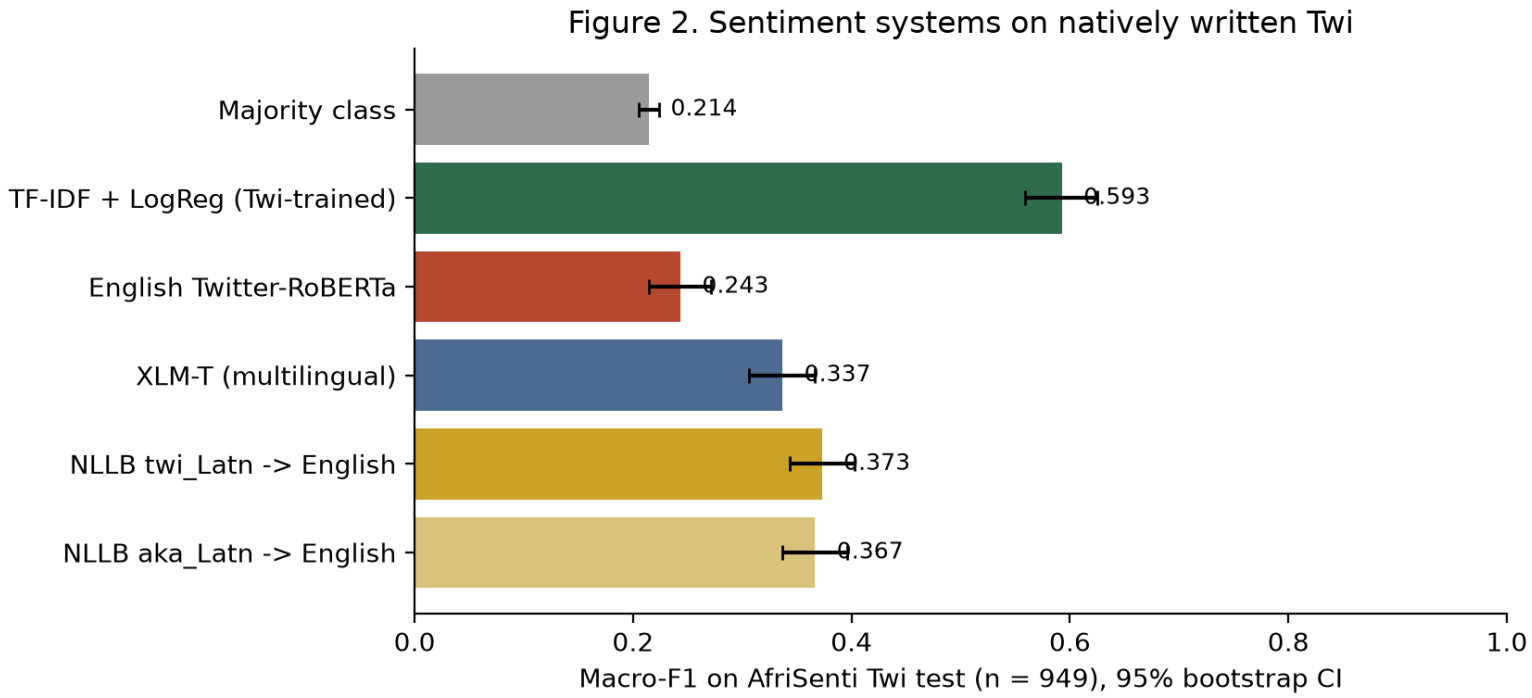
Multilingual: XLM-T, zero-shot (trained on 8 languages, none of them Twi).

Translate then classify: NLLB-200 Twi to English, then the English model. Run twice, declaring the source as twi_Latn and as aka_Latn.

Metrics: macro-F1 with 95% bootstrap CI, McNemar against the baseline, and the share of positive or negative tweets called neutral.

RQ3 · results

A small model trained on Twi beats every off-the-shelf system tested



Baseline 0.593

XLM-T 0.337

NLLB twi 0.373

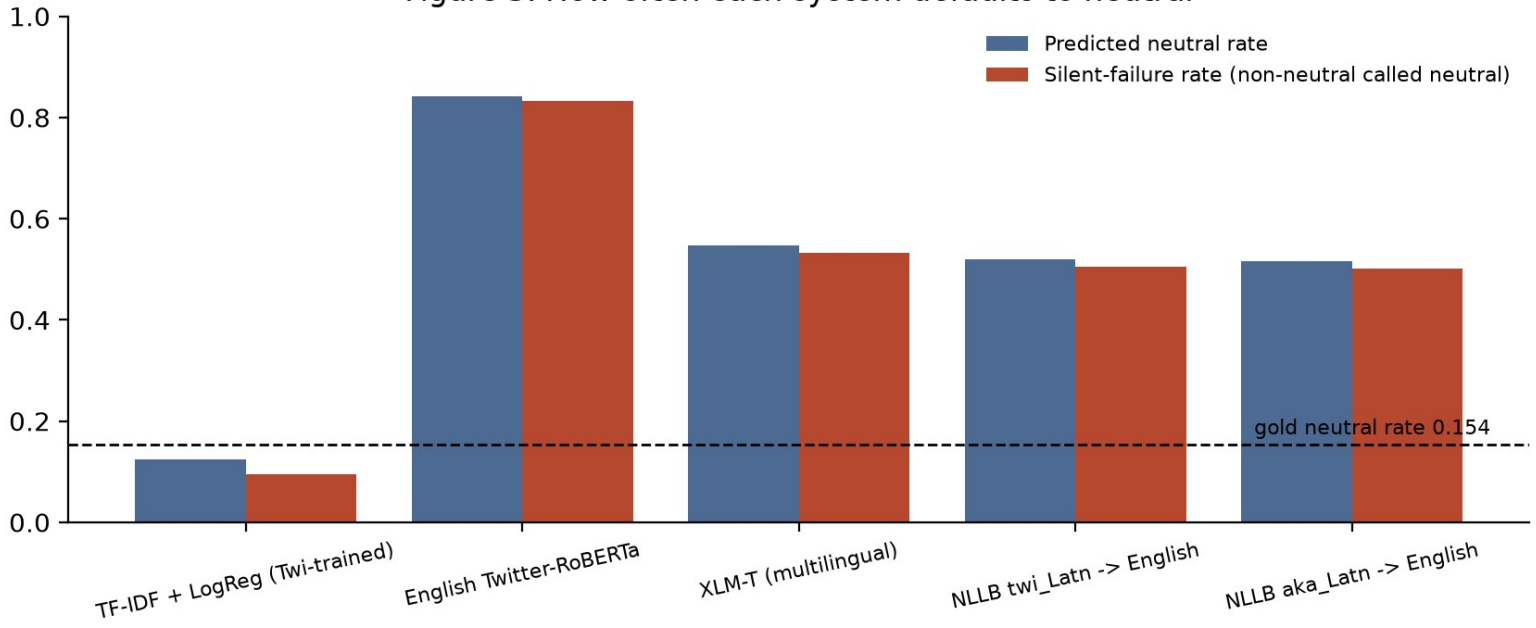
English-only 0.243

Majority 0.214

RQ3 · the silent failure

English-only tools turn Twi into neutral

Figure 3. How often each system defaults to neutral



Gold neutral rate: 15.4%

English-only predicts neutral 84.2% of the time

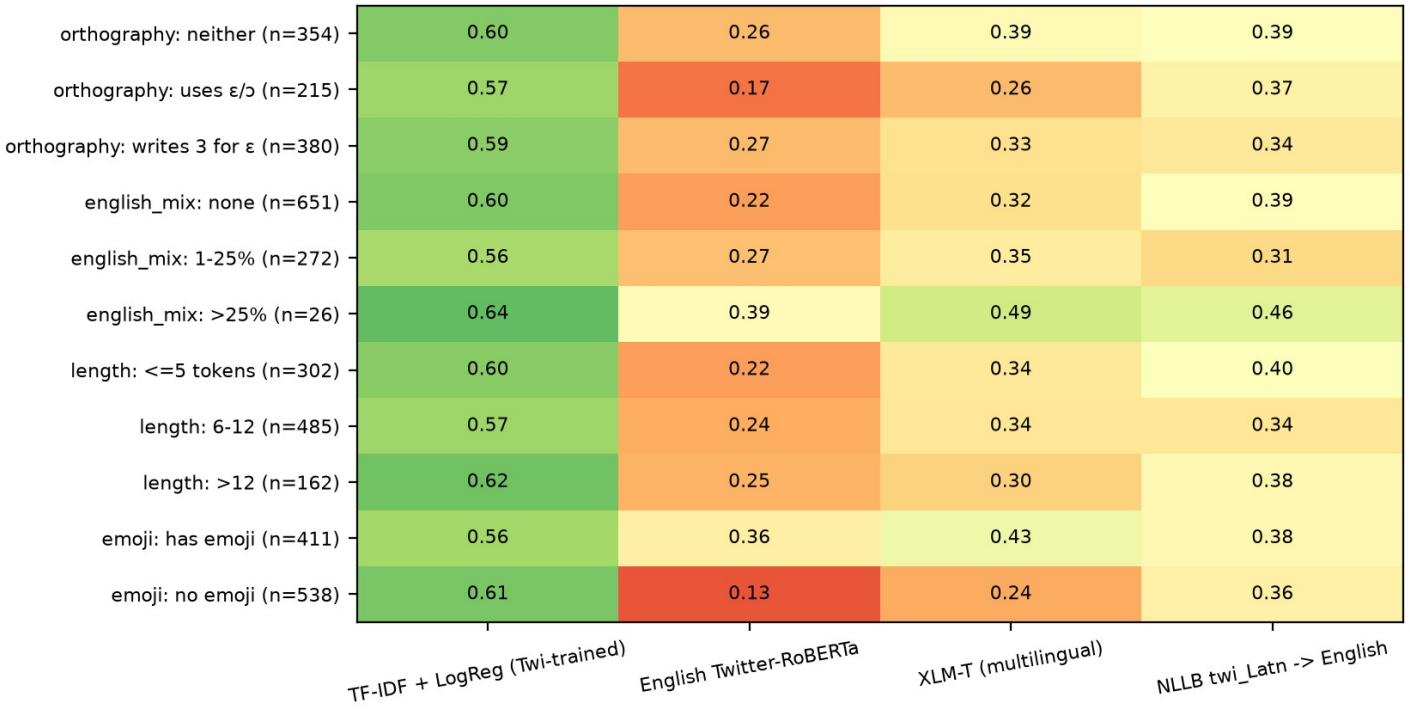
and calls 83.3% of positive or negative tweets neutral.

Its output carries no warning that anything went wrong.

Error analysis

Where the systems break

Figure 5. Macro-F1 by slice of the test split



Changing only the declared source (twi_Latn vs aka_Latn) changes the translation of 829 of 949 tweets and the final label of 243.

Tested by removing emoji: 98 of the English model's 125 non-neutral calls on emoji tweets turn neutral.

Rewriting 3 as ɛ stops NLLB copying it but does not significantly help the pipeline ($p = 0.76$); it does help XLM-T ($p = 0.027$).

132 tweets defeat the baseline, XLM-T and NLLB alike.

Limitations

"Not reported" means not verifiable from the public record, not that the practice was absent.

I speak Twi but did not grade translations, so translation results are about sentiment lost, not translation errors.

One labelled Twi social corpus exists, with kappa 0.51. Every result inherits its label noise.

Google Translate was not tested (no API key). Ewe and Ga could not be evaluated: no data.

Conclusion

What Ghanaian social media mining needs next

Twi capability claims rest on text translated into Twi. The only natively written labelled corpus is 4,818 tweets, and off-the-shelf tools fail on it in a way that looks like neutrality.

On that corpus, a small model trained on Twi scored far above every off-the-shelf system tested.

Needed: a Ga corpus, a larger natively written Twi corpus, a convention for naming the variety, and agreement scores reported in comparable form.

Code, data manifests and notebooks: https://github.com/osamabinIaggin/dcit316-final-project